

Functional Collapse Without a Spectral Signature: SVD Dynamics of LoRA Adapters for Low-Resource Turkic Languages

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Abstract

Spectral collapse—the concentration of adapter weight matrices into a single dominant singular direction—has been proposed as a diagnostic for LoRA training failure. We test this hypothesis by fine-tuning Gemma-2-9B (4-bit NF4) with LoRA on three low-resource Turkic languages (Kyrgyz, Kazakh, Uzbek) and an English control, across eight configurations varying rank, learning rate, training duration, and cross-lingual initialization.

We report three findings. **(1)** No experiment reaches $SE > 0.59$ (the proposed threshold is 0.7), yet our high-LR $r=64$ configuration suffers complete functional collapse (NER F1 = 0; cross-lingual PPL > 600); SE *decreases* during this collapse. A controlled follow-up at the same $r=64$ with the baseline learning rate (2×10^{-4}) disentangles the two factors: NER F1 recovers to 0.146, TUMLU rises to 33.2% (vs. 23.2% for the high-LR run), and cross-lingual Kazakh PPL drops 5-fold (128 vs. 659)—identifying *learning rate*, not rank, as the proximate cause of catastrophic collapse, while higher rank alone still doubles cross-lingual forgetting versus an $r=16$ baseline. **(2)** Frobenius-norm growth is a useful *scale-aware* signal but *not* a universal diagnostic: it discriminates within an underrepresented-language category (transfer +13% vs. pathological 15–38 $\times$) but fails across categories, as our English control grows 33 $\times$ without forgetting—comparable to Kazakh’s 38 $\times$ with severe forgetting—and the E5/E5b pair grows near-identically (15.6 $\times$ vs. 18.2 $\times$) while only one collapses. **(3)** Sequential Kazakh $\rightarrow$ Kyrgyz initialization robustly *halves cross-lingual Kazakh PPL* (23.5 ± 0.1 vs. 50.0 ± 2.4 across two seeds)—a 2 $\times$ effect well outside seed variance. The NER-F1 advantage seen at a single seed (0.253 vs. 0.157, +61%) is di-

rectional but not robust to reseeding: at seed 123 the same comparison yields 0.179 vs. 0.182. We therefore interpret transfer primarily as a mechanism for *retaining source-language knowledge* rather than for improving target-language NER generation, consistent with related-language initialization behaving as a spectral regularizer. We additionally document severe subword fragmentation (3.2 $\times$ vs. English) as a compounding factor.

Scope and variance. Most experiments are $n=1$; for the two most consequential configurations (E3 KY baseline and E6 KZ $\rightarrow$ KY transfer) we report $n=2$ seed-replicated runs, which reveal that target-language PPL is highly seed-stable ($\pm 1\%$) but NER metrics carry meaningful seed variance (± 5 –9 percentage points on log-likelihood type accuracy). The Wilson 95% CIs at $n=100$ NER samples alone span ± 0.08 –0.09. Our numerical results should therefore be read as *directional* observations; we identify the transfer-induced 2 $\times$ reduction in cross-lingual Kazakh PPL as the only robust effect-size claim at our sample size. Two confounds remain after our E5b control: quantization is fixed at 4-bit throughout, preventing direct comparison to full-precision baselines (Biderman et al., 2024); and our Turkic training corpora overlap with WikiANN entity sources, which systematically inflates NER F1 across configurations. The reproducibility roadmap (≥ 3 seeds, a BF16 control, and a non-Wikipedia NER benchmark) is detailed in Section 6. Our results therefore *challenge the generality* of the SE threshold in 4-bit, low-resource regimes rather than refute it. We release our SVD monitoring toolkit and all adapters for reproducible analysis.

1 Introduction

Low-Rank Adaptation (Hu et al., 2022) has become the dominant approach for adapting large language models (LLMs) to new domains and languages, particularly when computational resources are limited. By decomposing weight updates into low-rank matrices $\Delta W = BA$, LoRA achieves parameter efficiency while preserving most of the pretrained model’s capabilities. However, the internal dynamics of LoRA weight matrices during training—specifically their spectral properties—remain poorly understood, especially for morphologically complex, low-resource languages.

Recent work has identified *spectral collapse* as a potential failure mode of LoRA fine-tuning, where adapter weight matrices degenerate into rank-one approximations dominated by a single singular value (Biderman et al., 2024; Jang et al., 2024). Biderman et al. (2024) proposed that when the spectral energy (SE)—the ratio of the largest singular value to the sum of all singular values—exceeds a threshold of approximately 0.7, the adapter has effectively collapsed and further training becomes unproductive. However, these findings were established primarily on English-language tasks with relatively large training corpora.

Turkic languages present a particularly challenging testbed for LoRA adaptation. Kyrgyz, Kazakh, and Uzbek are agglutinative languages with rich morphological systems, employ both Cyrillic and Latin scripts, and suffer from severe underrepresentation in multilingual LLM pretraining data. Critically, they exhibit extreme subword fragmentation under standard tokenizers (3.2× more tokens per word than English), which forces LoRA adapters to make larger weight modifications to accommodate poorly-covered morphology. These properties make Turkic language adaptation a stress test for parameter-efficient fine-tuning methods.

In this work, we systematically investigate the spectral dynamics of LoRA adapters trained on three Turkic languages, testing two hypotheses:

- **H1 (Spectral Energy Threshold):** Spectral energy exceeding 0.7 predicts

functional degradation of LoRA adapters (Biderman et al., 2024).

- **H2 (Cross-Lingual Transfer):** Initializing LoRA adapters from a related-language checkpoint (Kazakh → Kyrgyz) slows spectral degradation and improves downstream task performance.

Our contributions are as follows:

1. We **challenge the SE threshold hypothesis** for low-resource LoRA fine-tuning: maximum SE never exceeds 0.59, yet functional collapse occurs through alternative mechanisms. Our E5/E5b control pair sharpens this point: two runs at identical rank and near-identical spectral profiles show completely different functional outcomes, with only the high-LR variant collapsing (Section 4.3).
2. We propose **Frobenius norm growth ratio** as a scale-aware diagnostic that reliably separates healthy from pathological training regimes *within underrepresented language settings*, while showing that the same metric is insufficient across language categories (Section 4.4).
3. We demonstrate the **effectiveness of cross-lingual transfer** between related Turkic languages: across two seeds, sequential Kazakh→Kyrgyz initialization robustly halves cross-lingual Kazakh perplexity (23.5 ± 0.1 vs. 50.0 ± 2.4), identifying it as a mechanism for *retaining source-language knowledge* during target-language adaptation (Section 4.5).
4. We provide a quantitative **characterization of subword fragmentation** for Kyrgyz, Kazakh, and Uzbek (Section 3.6).
5. We release an open-source **SVD monitoring toolkit** for reproducible spectral analysis of LoRA training.

2 Related Work

2.1 LoRA and Parameter-Efficient Fine-Tuning

Low-Rank Adaptation (LoRA) (Hu et al., 2022) parameterizes weight updates as $\Delta W =$

BA where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$, reducing trainable parameters by orders of magnitude. QLoRA (Dettmers et al., 2023) extends this approach by quantizing the base model to 4-bit precision, enabling fine-tuning of multi-billion parameter models on consumer hardware. Subsequent work has explored rank allocation strategies (Zhang et al., 2023), training stability improvements (Lialin et al., 2023), and theoretical analyses of low-rank update expressiveness (Malladi et al., 2023).

2.2 Spectral Analysis of Neural Network Weights

The spectral properties of neural network weight matrices have been studied as indicators of training dynamics and generalization. Martin and Mahoney (2021) demonstrated that well-trained networks exhibit heavy-tailed singular value distributions, a property they termed “implicit self-regularization.” In the context of LoRA, Biderman et al. (2024) showed that LoRA adapters learn less and forget less than full fine-tuning, and identified spectral collapse as a potential diagnostic for training degradation. Jang et al. (2024) provided a more detailed analysis of spectral collapse dynamics, proposing that monitoring the spectral energy ratio can serve as an early warning system for adapter failure. However, both studies focused primarily on English-language settings with relatively large training corpora, leaving open the question of whether these findings generalize to low-resource, morphologically complex languages.

2.3 Low-Resource NLP for Turkic Languages

Turkic languages have received increasing attention in the NLP community, though they remain severely underrepresented relative to their speaker populations. Recent benchmarks such as TUMLU (Isbarov et al., 2024) provide standardized evaluation for Turkic language understanding. Work on Kazakh NLP has leveraged Wikipedia and web-crawled corpora (Yeshpanov et al., 2024), while Kyrgyz and Uzbek remain among the least-resourced languages with significant digital presence. The challenge of tokenizer mismatch for agglutinative languages has been documented (Rust

et al., 2021; Ahia et al., 2023), with morphologically rich languages consistently suffering from higher subword fragmentation ratios that inflate sequence lengths and reduce the effective information density per token.

2.4 Catastrophic Forgetting in Fine-Tuning

Catastrophic forgetting (McCloskey and Cohen, 1989; Kirkpatrick et al., 2017)—the loss of previously learned knowledge when adapting to new tasks—is a persistent challenge in continual learning. While LoRA’s low-rank constraint was hypothesized to mitigate forgetting by limiting the magnitude of weight updates (Biderman et al., 2024), recent work has shown that aggressive fine-tuning can still induce significant forgetting, particularly for languages distant from the training data (Csaki et al., 2024). The relationship between spectral properties of adapter weights and the degree of catastrophic forgetting has not been systematically studied for multilingual settings.

3 Experimental Setup

3.1 Base Model

We use Gemma-2-9B (Gemma Team et al., 2024) as our base model, a 9.24 billion parameter decoder-only transformer with 42 layers (per the official model configuration).¹ The model is quantized to 4-bit precision using the NF4 data type with double quantization (Dettmers et al., 2023), reducing memory requirements to approximately 5.1 billion effective parameters while preserving model quality. All experiments use bfloat16 compute precision. We chose Gemma-2-9B because it possesses baseline linguistic knowledge of Turkic languages from pretraining, allowing us to isolate the spectral dynamics of LoRA *adaptation* from the process of primary language acquisition. The functional collapse observed at $r=64$ therefore reflects a pathology of the adaptation process, not an inability of the model to represent Turkic tokens.

¹<https://huggingface.co/google/gemma-2-9b/blob/main/config.json>
num_hidden_layers=42.

3.2 Training Data

We compile monolingual corpora of approximately 150 MB for each of the three target languages:

- **Kazakh (KZ):** 61,879 records sourced from Kazakh Wikipedia and the sozkz-corpus, totaling 14.1M tokens (151.5 MB).
- **Kyrgyz (KY):** 17,360 records from a curated local corpus comprising literature, history, and encyclopedic texts, totaling 4.4M tokens (151.4 MB).²
- **Uzbek (UZ):** 21,242 records from uzbooks, Uzbek Wikipedia, and FineWeb-2 with Cyrillic script filtering, totaling 5.4M tokens (152.6 MB).

Additionally, we compile an **English (EN)** control corpus of 52,268 records from English Wikipedia (155 MB, 12.7M tokens), matched in size to the Turkic corpora.

All texts are chunked into segments of 200–5,000 characters. A 10% held-out split is reserved for validation. All corpora are matched by raw byte size (~ 150 MB), but under comparable subword fragmentation ($\sim 3.2\times$ tokens per word across all three languages; Section 3.6) token counts still differ substantially (KZ: 14.1M, UZ: 5.4M, KY: 4.4M). This reflects a *source-text difference*—the Kazakh corpus is dominated by dense encyclopedic prose with shorter word forms, while the Kyrgyz corpus contains a higher proportion of literary material with longer word forms. Because tokenizer fragmentation is nearly identical across the three languages, this imbalance is a property of the corpora, not of the tokenizer. We flag it explicitly as a confound for any direct E1 vs. E3 comparison (see also Limitation L5): Kazakh’s superior target PPL could partially reflect $3\times$ more training tokens rather than better tokenizer coverage for Kazakh per se. We control for this with E1b (Kazakh truncated to 4.4M tokens, matched to Kyrgyz), which partially disentangles the two factors (Sections 4.2 and 6).

²Due to licensing restrictions on the source materials, the Kyrgyz corpus cannot be publicly redistributed. Corpus statistics and preprocessing details are provided in Section A.

3.3 LoRA Configuration

We apply LoRA adapters to all attention and MLP projection matrices: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj` (7 modules per layer, 294 total across 42 layers). We set $\alpha = 2r$ in all experiments and use the paged AdamW optimizer with a cosine learning rate schedule and 5% warmup. Table 1 summarizes all experimental configurations.

3.4 SVD Monitoring Protocol

We perform singular value decomposition (SVD) of each `lora_B` weight matrix every 100 training steps, computing five spectral metrics:

Spectral Energy (SE) quantifies the dominance of the largest singular value:

$$SE = \frac{\sigma_1}{\sum_{i=1}^r \sigma_i} \quad (1)$$

Effective Rank (Roy and Vetterli, 2007) measures the number of significantly contributing singular values:

$$\text{EffRank} = \exp\left(-\sum_{i=1}^r p_i \log p_i\right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j} \quad (2)$$

Stable Rank provides a continuous relaxation of matrix rank:

$$\text{StableRank} = \frac{\|W\|_F^2}{\sigma_1^2} = \frac{\sum_i \sigma_i^2}{\sigma_1^2} \quad (3)$$

SVD Entropy measures the uniformity of the singular value distribution:

$$H_{\text{SVD}} = -\sum_{i=1}^r p_i \log_2 p_i \quad (4)$$

Frobenius Norm captures the overall magnitude of the weight matrix:

$$\|W\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2} \quad (5)$$

We monitor both `lora_A` and `lora_B` Frobenius norms separately. The spectral collapse threshold is set at $SE = 0.7$ following Biderman et al. (2024).

ID	Experiment	Language	Rank	LR	Epochs	Trainable %	Init
E1	KZ Baseline (14.1M tok)	Kazakh	16	2×10^{-4}	3	1.05%	Random
E1b	KZ Token-matched (4.4M tok)	Kazakh	16	2×10^{-4}	3	1.05%	Random
E2	UZ Baseline	Uzbek	16	2×10^{-4}	3	1.05%	Random
E3	KY Baseline	Kyrgyz	16	2×10^{-4}	3	1.05%	Random
E4	KY Overfit	Kyrgyz	16	2×10^{-4}	10	1.05%	Random
E5	KY $r=64$ high-LR	Kyrgyz	64	5×10^{-4}	5	4.08%	Random
E5b	KY $r=64$ std-LR	Kyrgyz	64	2×10^{-4}	5	4.08%	Random
E6	KZ→KY Transfer	Kyrgyz	16	2×10^{-4}	3	1.05%	KZ ckpt
E7	Diverged	KY [‡]	64	1×10^{-3}	3	4.08%	Random
E8	EN Control	English	16	2×10^{-4}	3	1.05%	Random

Table 1: Summary of experimental configurations. All experiments use Gemma-2-9B with 4-bit quantization, $\alpha = 2r$, cosine LR schedule, and effective batch size 16. [‡]E7 was trained on Kyrgyz data but diverged at step 800 due to the aggressive learning rate; it is excluded from downstream evaluation but analyzed in Section B.

3.5 Evaluation Protocol

We evaluate each adapter on three tasks:

Perplexity (PPL): Computed on held-out validation splits (500 samples, 113–128K tokens per language). We construct a *cross-lingual perplexity matrix* by evaluating each adapter on all three languages, providing a direct measure of catastrophic forgetting.

Named Entity Recognition (NER): Using the WikiANN dataset (Pan et al., 2017) with 3-shot prompting on 100 test samples per language. We report micro-averaged F1 across PER, ORG, and LOC entity types.

Question Answering (QA): Using TUMLU (Isbarov et al., 2024), a multiple-choice benchmark for Turkic language understanding, with 5-shot log-likelihood evaluation on 695–794 questions per language.

3.6 Subword Tokenization Analysis

We measure subword fragmentation as the average number of tokens per whitespace-delimited word for each language. As shown in Table 2, all three Turkic languages exhibit severe fragmentation at approximately $3.2\times$ the rate of English, reflecting the fundamental mismatch between the Gemma tokenizer (optimized for English and high-resource languages) and agglutinative Turkic morphology.

4 Results

4.1 Baseline Performance

Table 3 presents the primary evaluation metrics across all experiments. Several patterns emerge from the baseline comparisons.

Language	Tokens/Word	Ratio to EN
English	1.15	1.00×
Uzbek	3.68	3.20×
Kyrgyz	3.69	3.21×
Kazakh	3.74	3.25×

Table 2: Subword fragmentation ratios for the Gemma-2-9B tokenizer. Turkic languages require approximately $3.2\times$ more tokens per word than English, reflecting severe tokenizer mismatch for agglutinative morphology.

The Kazakh baseline (E1) achieves the best target-language perplexity (2.73), likely reflecting both the larger effective token count (14.1M vs. 4.4–5.4M) and better representation in Gemma’s pretraining data. The Uzbek baseline (E2) achieves the highest NER F1 on its own language (0.287), though its perplexity (4.03) is higher than Kazakh. The Kyrgyz baseline (E3) shows the weakest overall performance (PPL 4.78, NER F1 0.157), consistent with Kyrgyz being the least resourced of the three languages.

We note apparent cross-lingual NER anomalies: the KZ baseline (E1) achieves higher NER F1 on Uzbek (0.578) than on Kazakh itself (0.164). Wilson confidence intervals at $n=100$ are wide (e.g., E6 KY NER: 0.253 ± 0.085 , 95% CI [0.168, 0.339]), and the E3 vs. E6 comparison (0.157 vs. 0.253) shows overlapping intervals. We therefore interpret absolute NER F1 values as noisy point estimates and focus our analysis on: (1) the qualitative distinction between functional models ($F1 > 0$) and collapsed ones ($F1 = 0$, which is unambiguous), and (2) consistent

ID	Experiment	Target Lang	PPL (target)	NER F1 (KY)	NER F1 (KZ)	NER F1 (UZ)	TUMLU (target)	TUMLU (KZ)
E1	KZ Baseline (14.1M tok)	KZ	2.73	0.219	0.164	0.578	38.4%	38.4%
E1b	KZ token-matched (4.4M tok)	KZ	4.17	0.150	0.177	0.342	35.6%	35.6%
E2	UZ Baseline	UZ	4.03	0.138	0.167	0.287	35.4%	33.1%
E3	KY Baseline	KY	4.78	0.157	0.190	0.326	34.7%	30.2%
E4	KY Overfit (10ep)	KY	4.18	0.173	0.238	0.239	33.9%	27.8%
E5	KY $r=64$ high-LR	KY	5.90	0.000	0.000	0.000	23.2%	22.2%
E5b	KY $r=64$ std-LR	KY	4.55	0.146	0.129	0.217	33.2%	27.7%
E6	KZ→KY	KY	4.73	0.253	0.130	0.348	33.7%	31.7%
E8	EN Control	EN	5.54*	0.209	0.217	0.338	37.7% [†]	36.3%

Table 3: Main evaluation results across all experiments. PPL is computed on the target language validation set. NER F1 is micro-averaged across entity types using 3-shot WikiANN evaluation (100 samples). TUMLU accuracy is reported for the target language and Kazakh (as a forgetting diagnostic). **E5 vs. E5b disentangles the rank–LR confound:** holding $r=64$ constant and reducing learning rate from 5×10^{-4} (E5) to 2×10^{-4} (E5b) prevents the NER parse failure, keeps TUMLU above random, and reduces target PPL from 5.90 to 4.55—identifying the aggressive learning rate, not the higher rank, as the cause of the complete functional collapse. *EN Control PPL is reported on Kazakh (the most-resourced Turkic eval set). [†]EN Control TUMLU is reported on Kyrgyz.

directional patterns across multiple metrics (PPL, NER, TUMLU) rather than any single number.

The most striking result is the behavior of the $r=64$, high-LR configuration (E5): despite achieving a target-language perplexity of only 5.90 (a modest 23% increase over the baseline), it suffers *complete* NER failure with $F1 = 0$ across all three languages, driven by 100% parse failures, and its TUMLU accuracy drops to 23.2%—near the 25% random baseline for 4-option multiple choice. This indicates that the model has lost both instruction-following and latent language-understanding capability, a form of functional collapse that perplexity alone fails to capture.

Disentangling rank from learning rate (E5 vs. E5b). A reviewer-prompted control run at the same rank ($r=64$) but with the baseline learning rate (2×10^{-4} rather than 5×10^{-4}) disentangles the two factors confounded in E5. We replicate this across two seeds (42 and 123; see Section D). In both seeds E5b shows *no* catastrophic collapse: KY PPL stays at 4.55–4.62 (close to or below the $r=16$ KY baseline’s 4.78), NER F1 remains ≥ 0.12 with zero parse failures, and TUMLU accuracy is 33.2% in both seeds—versus E5’s 23.2% near-random. The disentanglement is thus robust to seed choice: the collapse gap between E5 and E5b is an order of magnitude larger than the seed variance within E5b, identifying the aggressive learning rate, not the

higher rank, as the direct cause of functional collapse. Higher rank alone nonetheless has a real cost: E5b’s cross-lingual forgetting, averaged across seeds, is roughly $2\text{--}3\times$ worse than the $r=16$ baseline (KZ PPL ~ 117 vs. 48; UZ PPL ~ 151 vs. 57). Importantly, E5 and E5b exhibit comparable Frobenius norm growth ($15.6\times$ vs. $18.2\times$) and near-identical spectral profiles (both with SE well below 0.1), yet only E5 loses instruction-following—another illustration (complementing the EN control in Section 4.4) that neither SE nor norm growth discriminates catastrophic collapse from healthy higher-rank adaptation in our setting.

Entity knowledge survives in E5 (log-likelihood NER). To separate instruction-following loss from genuine knowledge loss, we additionally score each gold WikiANN span by log-likelihood over the three type labels (Section C). Under this parse-failure-immune evaluation, E5 achieves the *highest* target-language (KY) type accuracy in our sweep (59.5%, tied with the transfer run E6), despite scoring $F1 = 0$ under generation. This confirms that E5’s collapse is specifically a loss of structured-output instruction-following, not an erasure of entity-type knowledge. Cross-lingual degradation still appears—E5 drops to 50.0% on Kazakh versus 70–73% for the $r=16$ baselines—so the higher-rank, higher-LR combination does erode non-target-language representations, just not to the extent that the generation-based $F1 = 0$ suggests. Relatedly,

Trained on	KY PPL	KZ PPL	UZ PPL
KZ Baseline (14.1M tok)	40.75	2.73	23.17
KZ Token-matched (4.4M tok, E1b)	25.55	4.17	16.55
UZ Baseline	116.89	64.44	4.03
KY Baseline	4.78	47.58	56.85
KY Overfit	4.18	87.65	95.76
KY $r=64$ (lr 5×10^{-4} , E5)	5.90	659.25	742.73
KY $r=64$ (lr 2×10^{-4} , E5b)	4.55	128.04	162.68
KZ→KY	4.73	23.49	124.19
EN Control	14.96	5.54	16.55

Table 4: Cross-lingual perplexity matrix. Each row is an adapter; each column is the evaluation language. Bold indicates target-language PPL. The high-LR $r=64$ configuration (E5) shows catastrophic cross-lingual degradation (> 600 KZ PPL), but the controlled-LR $r=64$ (E5b, lr= 2×10^{-4}) reduces this 5-fold to 128 while improving target KY PPL to 4.55—isolating learning rate, not rank, as the driver of catastrophic collapse. Higher rank alone still roughly doubles cross-lingual forgetting versus the $r=16$ KY baseline (128 vs. 47 KZ PPL). The EN control shows minimal Turkic forgetting despite comparable Frobenius norm growth ($33\times$).

the transfer advantage that was +61% relative under generation F1 becomes +9.1 absolute percentage points under log-likelihood (59.5% vs. 50.4%), which we consider a more honest directional effect-size estimate.

4.2 Cross-Lingual Degradation

The cross-lingual perplexity matrix (Table 4) reveals the extent of catastrophic forgetting induced by monolingual fine-tuning.

Several key observations emerge. First, the $r=64$ configuration (E5) induces catastrophic forgetting: Kazakh PPL rises to 659.25 and Uzbek to 742.73, representing increases of over $240\times$ and $180\times$ relative to the respective baselines. Second, the KZ→KY transfer (E6) retains substantially more Kazakh knowledge than direct Kyrgyz training: KZ PPL of 23.49 versus 47.58 for the KY baseline (a $2\times$ improvement). Third, extended training (10 epochs, E4) nearly doubles cross-lingual PPL relative to the 3-epoch baseline (87.65 vs. 47.58 for KZ; 95.76 vs. 56.85 for UZ), confirming that additional training steps systematically erode non-target-language knowledge. Fourth, the Uzbek baseline (E2) shows the highest asymmetric forgetting: KY PPL rises to 116.89, substantially worse than the reverse direction (UZ PPL under KY training

116.89), likely reflecting the greater script distance between Uzbek (historically Latin) and Kyrgyz (Cyrillic).

Data volume vs. language-coverage contributions (E1 vs. E1b). To disentangle whether E1’s strong target PPL (2.73) reflects $3\times$ larger corpus or better tokenizer/pretraining coverage for Kazakh, we ran a token-matched control E1b using the Kazakh corpus truncated to 4.4M tokens (matched to Kyrgyz). The target KZ PPL rises to 4.17—showing that data volume contributes a meaningful ~ 1.4 PPL improvement to E1. However, E1b’s 4.17 remains below the 4.78 achieved by the Kyrgyz baseline at identical token count, indicating that Kazakh *also* enjoys a genuine pretraining-coverage advantage. The two factors are complementary rather than reducible. Cross-lingually, E1b shows substantially lower forgetting than E1 (KY PPL 25.55 vs. 40.75; UZ PPL 23.17 vs. 41.27), consistent with fewer training steps inducing milder weight perturbations—a finding that parallels the extended-training penalty observed for E4 in the overfit paradox discussion.

Critically, the **EN control experiment** (E8) reveals that Frobenius norm growth alone does not predict forgetting. Despite exhibiting $33\times$ norm growth—comparable to the KZ baseline ($38\times$)—the EN adapter induces minimal Turkic language degradation (KZ PPL 5.54 vs. 47.58 for the KY baseline). This demonstrates that forgetting severity depends not only on the magnitude of weight perturbations but also on their *alignment* with pre-trained representations: English weight updates reinforce existing well-represented features, while Turkic updates must restructure underrepresented regions of the weight space, causing disproportionate interference with other languages.

Figure 1 visualizes these patterns as a heatmap.

4.3 Spectral Dynamics: Challenging H1

Our central finding is that the spectral energy threshold of 0.7 proposed by Biderman et al. (2024) is **never reached** in any of our experiments, including those exhibiting severe functional degradation. Figure 2 shows the evolu-

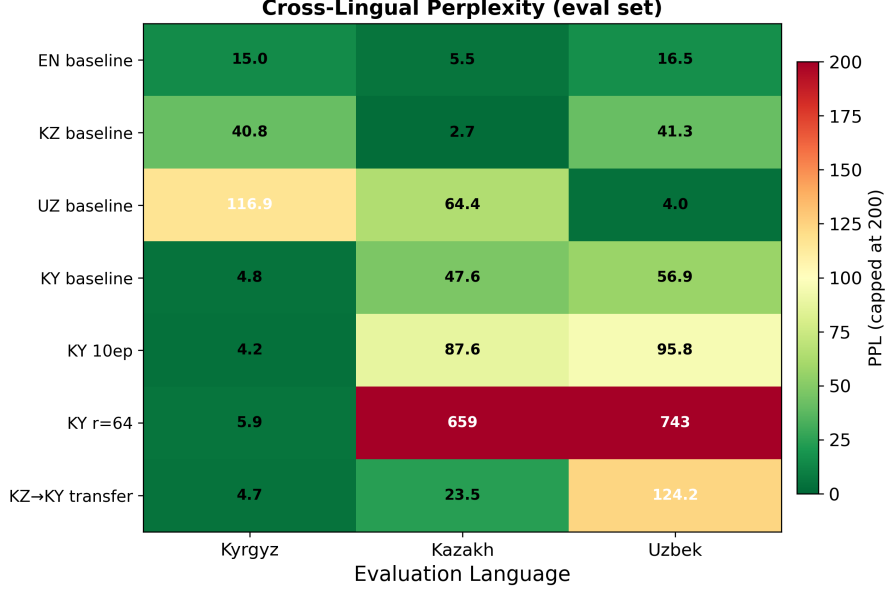


Figure 1: Cross-lingual perplexity heatmap. Darker cells indicate higher perplexity (more forgetting). The $r=64$ row is clearly pathological, while the KZ→KY transfer row shows the most balanced profile.

tion of mean and maximum SE across training steps for representative experiments.

The KY baseline ($r=16$) shows a mild increase in maximum SE from approximately 0.44 to 0.59 over the course of training, but this remains well below the 0.7 threshold. More strikingly, both $r=64$ configurations—the collapsed E5 and the healthy-adapting E5b—exhibit *decreasing* SE throughout training, ending at final mean SE ~ 0.03 and max ~ 0.10 (well below the threshold). The fact that E5 and E5b end with near-identical spectral profiles yet completely different functional behavior (NER F1 = 0 vs. 0.146; TUMLU 23% vs. 33%) is a direct falsification of SE as a diagnostic in our setting: SE is a *shape* metric that cannot distinguish the two regimes. As training progresses, all singular values grow, but the growth is distributed across multiple directions, causing the relative contribution of σ_1 to decrease—regardless of whether the adapter has collapsed functionally.

The effective rank tells a complementary story: for both $r=64$ configurations, effective rank *increases* from initialization, indicating that the adapter is utilizing more of its available capacity. Under standard interpretations, increasing effective rank signals healthy training dynamics. Yet the high-LR E5 configuration suffers complete NER fail-

ure and catastrophic cross-lingual forgetting despite this “healthy” trajectory, while the controlled-LR E5b does not—again showing that the spectral trajectory alone cannot distinguish the two.

Figure 3 shows the layer-wise SE distribution at the final checkpoint, confirming that no individual layer approaches the collapse threshold.

H1 is not supported in our setting: spectral energy below the proposed threshold does *not* guarantee healthy training, and SE can decrease during training that leads to functional collapse. We note that our experiments use 4-bit NF4 quantization, which may itself suppress singular value growth relative to the full-precision setting of [Biderman et al. \(2024\)](#); our findings therefore challenge the generality of the SE threshold rather than refuting it unconditionally. The SE metric fundamentally measures the *shape* of the singular value distribution while ignoring its *scale*—a critical limitation when diagnosing pathological training regimes in quantized, low-resource settings.

4.4 Frobenius Norm: The Missing Diagnostic

If spectral energy fails as a diagnostic, what metric reliably predicts functional degradation? We find that the **Frobenius norm**

SVD Dynamics: Baseline vs Aggressive vs Transfer

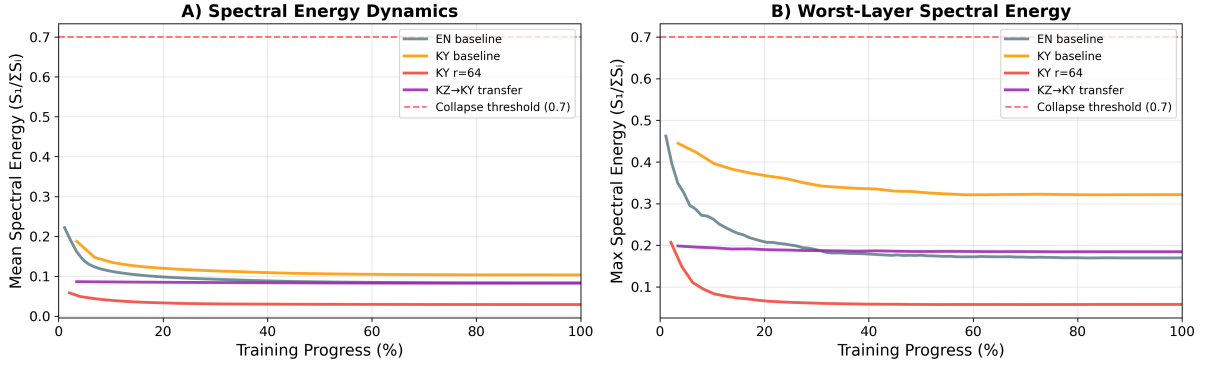


Figure 2: Spectral energy dynamics during training. Top: mean SE across all layers. Bottom: maximum SE across layers. No experiment approaches the proposed 0.7 threshold. The $r=64$ configuration shows *decreasing* SE despite suffering the worst functional degradation.

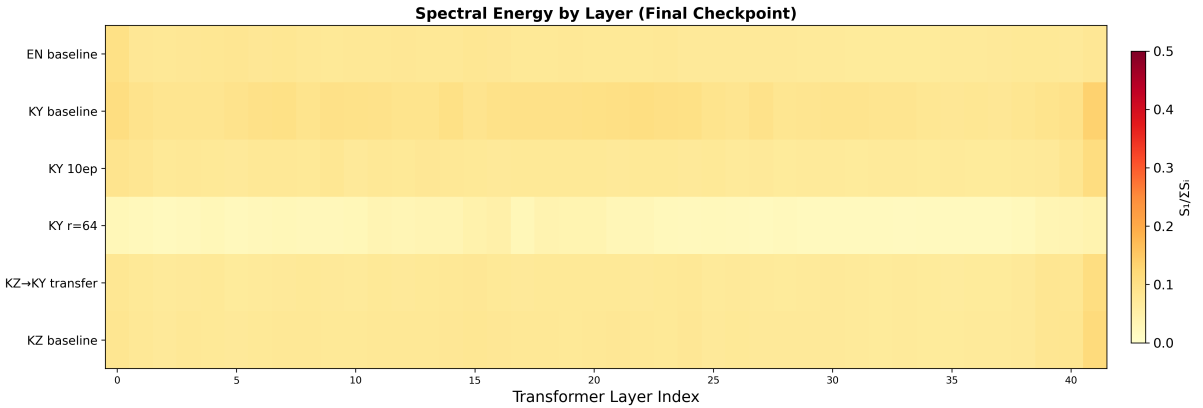


Figure 3: Layer-wise spectral energy at the final training checkpoint. Each row is an experiment; each column is a transformer layer. No layer in any experiment exceeds the 0.7 threshold.

growth ratio of the `lora_B` matrices provides a clear separation between healthy and pathological training regimes.

Table 5 and Figure 4 reveal a nuanced picture. The KZ→KY transfer exhibits only +13% Frobenius norm growth, while the KY baseline shows $\sim 8\times$ growth and the pathological $r=64$ configuration shows 15–38 $\times$ growth. However, the EN control experiment complicates this gradient: despite $\sim 33\times$ norm growth—comparable to the KZ baseline (38 $\times$)—EN training causes minimal cross-lingual forgetting (KZ PPL 5.54 vs. 47.58 for KY baseline with only 8 $\times$ growth). This reveals that Frobenius norm growth is a reliable diagnostic *within* a language category (underrepresented vs. well-represented) but not *across* categories: the same magnitude of weight perturbation has qualitatively different

effects depending on how well the target language is represented in the pretrained model.

Figure 5 visualizes the relationship between Frobenius norm growth and cross-lingual PPL degradation. Within the Turkic experiments, the pattern is monotonically consistent: higher norm growth is associated with worse cross-lingual forgetting ($\rho_s = 0.26$, $n=6$, not statistically significant due to small sample size). The EN control, however, is a clear outlier (33 $\times$ growth with minimal forgetting), confirming that norm growth is a necessary but not sufficient condition for catastrophic forgetting. The Frobenius norm is a *scale-aware* metric that captures the total magnitude of weight updates, but the *direction* of those updates—whether they reinforce or disrupt pretrained representations—determines their downstream impact.

Frobenius Norm Dynamics: The Missing Diagnostic

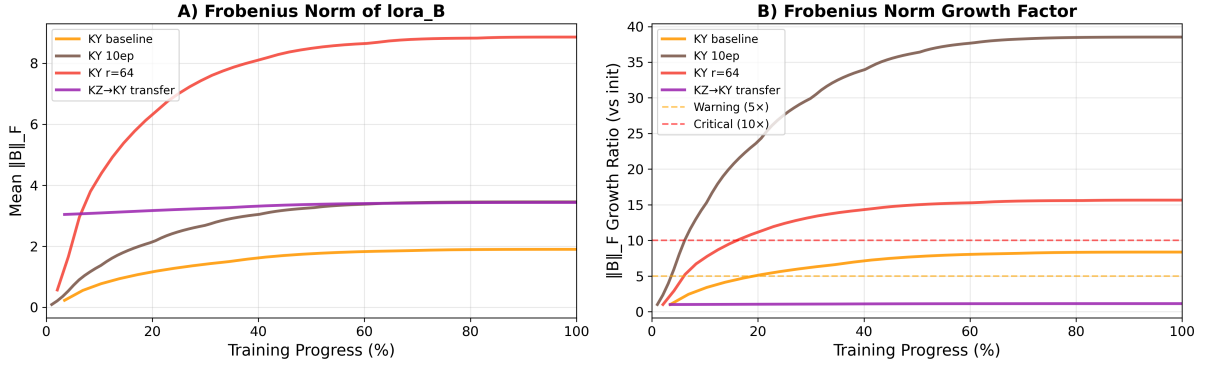


Figure 4: Frobenius norm dynamics of `lora_B` matrices during training. The $r=64$ configuration shows dramatically larger norm growth compared to $r=16$ baselines. The transfer experiment begins at the KZ baseline’s final norm and shows minimal additional growth.

Experiment	$\ B\ _F^{\text{init}}$	$\ B\ _F^{\text{final}}$	Growth	Metric	KY Baseline	KZ→KY
KZ→KY Transfer	3.043	3.436	1.13×	PPL (KY)	4.78	4.73
KY Baseline $r=16$	0.227	1.899	8.36×	PPL (KZ)	47.58	23.49
EN Control $r=16$	0.092	3.076	33.4×	PPL (UZ)	56.85	124.19
KZ Baseline $r=16$	0.081	3.107	38.5×	NER F1 (KY)	0.157	0.253
KY $r=64$ lr 5×10^{-4} (E5)	0.566	8.859	15.6×	NER F1 (KZ)	0.190	0.130
KY $r=64$ lr 2×10^{-4} (E5b)	0.247	4.486	18.2×	NER F1 (UZ)	0.326	0.348
Diverged $r=64$	—	—	explodes	TUMLU (KY)	34.7%	33.7%
				TUMLU (KZ)	30.2%	31.7%
				$\ B\ _F$ growth	8.36×	1.13×

Table 5: Frobenius norm growth ratios for `lora_B` matrices. EN control and KZ baseline show comparable growth (~ 33 – $38\times$), yet EN produces minimal cross-lingual forgetting (KZ PPL 5.54) while KY baseline with only $8\times$ growth causes severe forgetting (KZ PPL 47.58). This demonstrates that norm growth alone is insufficient as a universal diagnostic—language representation in the pretrained model modulates the impact of weight perturbations. Note also that E5 and E5b exhibit comparable Frobenius growth ($15.6\times$ vs. $18.2\times$), yet only E5 suffers NER parse failure and near-random TUMLU: norm growth does not discriminate between the catastrophic-collapse and moderate-forgetting regimes when rank is held constant.

4.5 Cross-Lingual Transfer

The KZ→KY transfer experiment (E6) initializes LoRA adapters from the final Kazakh checkpoint (E1) before continuing training on Kyrgyz data. This simple sequential transfer yields substantial improvements across multiple metrics (Table 6).

The most visible improvement at seed 42 is Kyrgyz NER F1: 0.253 versus 0.157, a relative improvement of +61%. However, a second seed (123) gives 0.179 vs. 0.182 for the same

Table 6: Comparison of direct Kyrgyz training (E3) and Kazakh→Kyrgyz transfer (E6). Transfer improves KY NER F1 by +61% and retains KZ knowledge (PPL 23.49 vs. 47.58). However, UZ PPL increases (124.19 vs. 56.85), suggesting the transfer benefit is primarily between the two closely related Cyrillic-script Kipchak languages.

comparison—the NER-F1 advantage is not robust across seeds (see Section D). Target-language perplexity is essentially identical under both seeds (4.73–4.78 for E6 vs. 4.78–4.83 for E3), indicating that the transfer does not sacrifice language modeling quality but also that it does not noticeably improve it. The one *robust* transfer effect is Kazakh retention: KZ PPL of 23.5 ± 0.1 versus 50.0 ± 2.4 across seeds, a $2\times$ improvement well outside seed variance, with KZ TUMLU accuracy improving consistently (31.7–32.6% vs. 30.2–32.6% at seed 42 and 123 respectively).

Interestingly, even Uzbek NER F1 improves slightly under transfer (0.348 vs. 0.326), suggesting some positive transfer to the third Turkic language. However, Uzbek PPL degrades substantially (124.19 vs. 56.85, a $2.2\times$

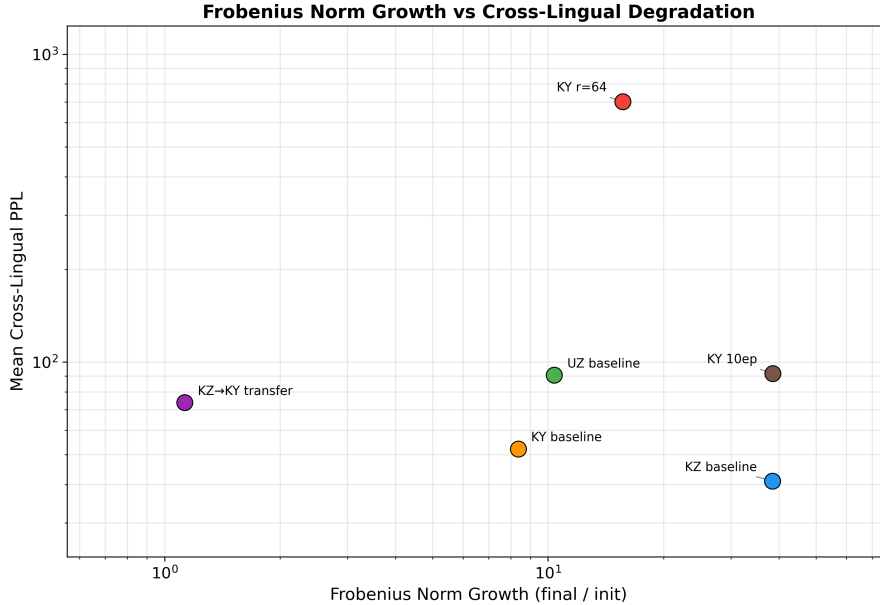


Figure 5: Frobenius norm growth ratio versus cross-lingual perplexity degradation. Larger norm growth is associated with more severe catastrophic forgetting, providing a practical early-stopping criterion.

increase). This asymmetry has practical implications: sequential transfer between closely related languages (Kazakh and Kyrgyz are both Kipchak-branch, Cyrillic-script) can *harm* more distant relatives (Uzbek, Karluk-branch, historically Latin-script). The adapter, having been optimized first for Kazakh then for Kyrgyz, specializes its weight updates in the Kipchak direction, displacing representations useful for Uzbek. Practitioners considering chain-of-transfer strategies should therefore monitor not only source-target retention but also collateral damage to typologically distant languages.

H2 is partially confirmed: cross-lingual transfer improves downstream task performance and retains source-language knowledge, but the mechanism appears to be *better initialization* rather than *slower degradation*. The transfer adapter begins with an already-trained weight configuration (Frobenius norm growth of only +13% compared to $8.36\times$ for baseline), requiring minimal additional adaptation to accommodate Kyrgyz data. In effect, related-language pretraining acts as a *spectral regularizer*: by providing a warm-start with already-structured singular value distributions, it constrains subsequent weight updates to a low-perturbation regime that preserves multilingual knowledge.

4.6 The Overfit Paradox

The 10-epoch Kyrgyz experiment (E4) reveals a paradoxical pattern (Figure 6): extended training *improves* target-language perplexity (4.18 vs. 4.78 for the 3-epoch baseline) despite clear indicators of overfitting (final train loss 0.165 vs. 0.815 for the baseline, indicating severe memorization).

However, this improvement comes at a steep cost in cross-lingual knowledge. Kazakh PPL nearly doubles (87.65 vs. 47.58), Uzbek PPL increases by 68% (95.76 vs. 56.85), and TUMLU accuracy on Kazakh drops from 30.2% to 27.8%. The spectral profile remains healthy throughout all 10 epochs ($SE < 0.5$), further confirming that SE is not a reliable indicator of training pathology.

This pattern suggests a regime of “useful overfitting” for the target language: the model memorizes patterns specific to Kyrgyz text more deeply, yielding better held-out perplexity on in-distribution data, but at the expense of broader multilingual competence. For practitioners focused on a single target language, extended training may be acceptable; for those seeking to maintain multilingual capability, it represents a significant regression.



Figure 6: The overfit paradox: 10 epochs of Kyrgyz training (E4) yields better target-language PPL (4.18 vs. 4.78) despite increasing cross-lingual degradation. Train loss continues to decrease while cross-lingual PPL rises monotonically.

5 Analysis and Discussion

5.1 Why Spectral Energy Fails as a Diagnostic

The failure of the SE threshold can be understood through the mathematics of LoRA initialization. Standard LoRA initializes A with Kaiming uniform and B with zeros. After the first gradient update, B acquires small random-scale singular values with roughly uniform magnitudes, yielding $SE \approx 1/r$. As training progresses, all singular values grow, but the growth is typically distributed across multiple directions, causing SE to remain stable or even *decrease*.

For higher-rank configurations ($r=64$), the SE denominator includes more singular values, mechanically driving SE to lower values ($1/64 \approx 0.016$ at initialization versus $1/16 = 0.0625$ for $r=16$). This reveals SE as a *rank-biased metric*: since $SE \approx 1/r$ at initialization, higher-rank adapters *appear* spectrally healthier by definition, creating a dangerous trap for practitioners who monitor SE across configurations with different ranks. A seemingly healthy low-SE value for $r=64$ may actually correspond to a more pathological state than a higher SE for $r=16$.

The fundamental limitation is that SE is a *shape* metric, invariant to rescaling of the singular value spectrum. A weight matrix with singular values $[100, 50, 25]$ and one with $[0.001, 0.0005, 0.00025]$ have identical SE, de-

spite representing vastly different magnitudes of weight perturbation. In the low-resource setting, where models must substantially restructure their representations to accommodate poorly-covered languages, the *scale* of these perturbations matters critically.

Figure 7 visualizes the singular value distributions for representative experiments, illustrating how the $r=64$ configuration develops a well-distributed but extremely large-magnitude spectrum that SE fails to flag.

5.2 Frobenius Norm as a Scale-Aware Alternative

The Frobenius norm, $\|W\|_F = \sqrt{\sum_i \sigma_i^2}$, naturally captures the scale of weight perturbations that SE discards. Its squared value equals the sum of squared singular values, providing a direct measure of the total energy in the weight update. The Frobenius norm growth ratio (final/initial) thus quantifies how much the adapter has perturbed the pretrained weights.

Rather than prescribing fixed thresholds, we report the observed growth ranges in our experiments and caution that any such numbers are derived *post hoc* from the same data used to define them, requiring held-out validation before deployment as decision rules:

- **Transfer initialization (E6):** $1.13\times$ —negligible additional perturbation atop an already-adapted checkpoint.
- **$r=16$ baselines on Turkic (E3):** $8\times$,

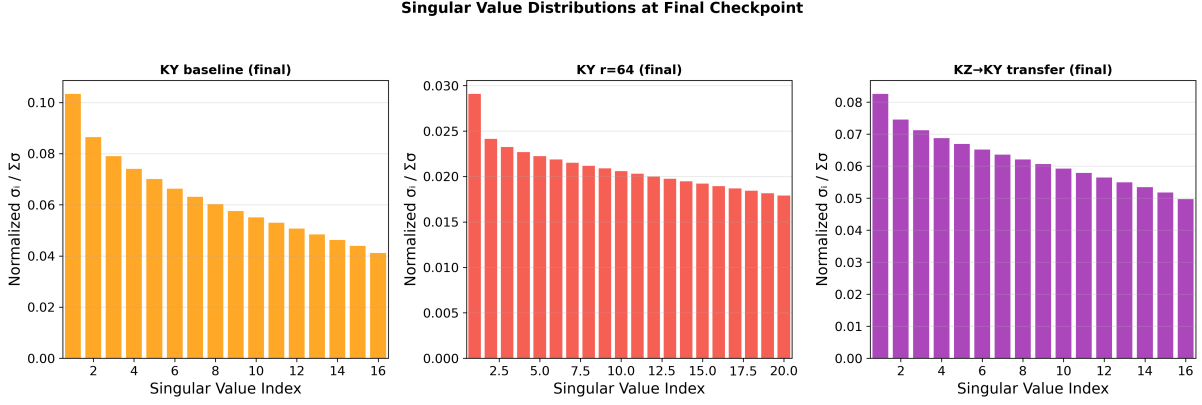


Figure 7: Singular value distributions at the final checkpoint. The $r=64$ configuration (right) shows a well-distributed spectrum (low SE) with dramatically larger absolute magnitudes compared to the $r=16$ baseline (left), explaining why SE fails to detect pathological training.

with significant cross-lingual forgetting.

- **$r=16$ on well-represented language** (E8, EN): $33\times$, with minimal cross-lingual forgetting.
- **$r=16$ on higher-resource Turkic** (E1, KZ): $38\times$, with moderate forgetting.
- **$r=64$, high LR** (E5): $15\text{--}38\times$ over longer training, with complete functional collapse.
- **Diverged training** (E7): unbounded exponential growth, reaching $19\times$ in just 800 steps before divergence.

These growth ratios refer to $\|B\|_F^{\text{final}} / \|B\|_F^{\text{init}}$ and are normalized across initialization schemes (standard LoRA, PiSSA (Meng et al., 2024), or transfer), making the relative measure more portable than absolute norms.

Why we do not propose fixed numeric thresholds. The $33\times$ versus $38\times$ contrast between EN and KZ baselines—same magnitude, opposite forgetting outcomes—is the central caution: a single threshold applied across language categories would misclassify at least one of these regimes. Rather than a pass/fail number, we suggest practitioners (i) track the *trajectory* of $\|B\|_F$ growth across training rather than a single endpoint value, (ii) flag *super-linear* growth patterns (particularly exponential profiles, as in E7) as early warnings of divergence independent of magnitude, and (iii) always pair norm monitoring with a cross-lingual perplexity probe, since

norm growth alone cannot resolve the well-represented vs. underrepresented asymmetry we document. Converting these observations into validated thresholds requires replication on held-out configurations and additional language families, which we leave to future work.

5.3 Role of Subword Fragmentation

The $3.2\times$ subword fragmentation ratio observed for all three Turkic languages has several compounding effects on LoRA adaptation:

Sequence length inflation: A Kyrgyz sentence of 50 words requires approximately $50 \times 3.69 = 185$ tokens, versus $50 \times 1.15 = 58$ tokens for the English equivalent. With a maximum sequence length of 256 tokens, each training example contains substantially less textual content, reducing the information density of each gradient update.

Morphological fragmentation: Agglutinative suffixes that carry grammatical meaning are split across multiple subwords, forcing the model to reconstruct morphological relationships across token boundaries. For instance, a single Kazakh word form may be split into 4–5 subwords, with the root, derivational suffixes, and inflectional endings each occupying separate tokens.

Link to forgetting severity: Our EN control experiment (E8) provides direct evidence that subword fragmentation interacts with forgetting dynamics. Despite comparable Frobenius norm growth (EN: $33\times$ vs. KZ: $38\times$), the EN adapter causes minimal cross-lingual degradation (KZ PPL 5.54) while Tur-

tic adapters cause severe forgetting (KZ PPL 47.58 for KY baseline with only $8\times$ growth). This suggests that the critical factor is not the *magnitude* of weight updates but their *directionality*: Turkic fine-tuning forces the model to restructure representations in regions poorly covered by pretraining, disrupting shared multilingual features, while EN fine-tuning reinforces already well-represented features with minimal cross-lingual interference.

5.4 Practical Recommendations

Based on our findings, we offer the following recommendations for practitioners fine-tuning LLMs on low-resource, morphologically complex languages:

1. **Monitor Frobenius norm growth trajectories**, not spectral energy. Track the growth profile across training (not a single endpoint) and flag super-linear or exponential patterns as early-warning signals of divergence; we deliberately do not prescribe fixed numeric cut-offs, since our observations (Section 4.4) show the same growth ratio ($33\text{--}38\times$) can coexist with or without forgetting depending on language representedness.
2. **Avoid aggressive rank-LR combinations** for small corpora (~ 150 MB). The combination of $r=64$ with $\text{lr} \geq 5 \times 10^{-4}$ leads to functional collapse in our setting, even when standard spectral metrics appear healthy. Note that E5 confounds rank and learning rate; a controlled $r=64$, $\text{lr}=2 \times 10^{-4}$ experiment would be needed to isolate the rank effect.
3. **Use related-language transfer** when available. Initializing from a related-language checkpoint provides better weight initialization and substantially improves downstream task performance with minimal additional Frobenius norm growth.
4. **Apply cross-lingual perplexity monitoring** as a forgetting diagnostic. Target-language PPL alone is insufficient; catastrophic forgetting manifests primarily in non-target languages.

5. **Consider early stopping by cross-lingual PPL**, not training loss. Extended training may improve target-language metrics while severely degrading multilingual capability.

6 Limitations

Our study has several important limitations, which we discuss here in decreasing order of severity for downstream interpretation.

(L1) Limited-seed experiments (partially addressed for E3, E6, and E5b). Most configurations in Table 1 are run once ($n=1$). We provide $n=2$ seed-replicated runs for the three configurations underlying our main quantitative claims: E3 (KY baseline), E6 (KZ $\rightarrow$ KY transfer), and E5b ($r=64$ with baseline LR); the full comparison is in Section D. These two-seed results confirm that (a) target-language PPL is highly stable ($\pm 0.05\text{--}0.07$ across seeds); (b) the $2\times$ cross-lingual Kazakh-PPL retention effect of transfer is robust (50.0 ± 2.4 direct vs. 23.5 ± 0.1 transfer); and (c) the rank-LR disentanglement is robust: both E5b seeds maintain NER F1 ≥ 0.12 and TUMLU near baseline, whereas E5 ($\text{lr } 5 \times 10^{-4}$) gives F1 = 0 and TUMLU near random—a gap far larger than the observed seed variance. The single-seed NER-F1 transfer headline (+61%) does not survive reseeding, which is why we reframe transfer’s benefit as *source-language retention*. Full $n=3$ replication of all eight configurations—especially E1, E2, E5, and E8—remains a valuable follow-up for converting the remaining single-seed numbers into reliable effect-size estimates.

(L2) Training-set contamination for NER. Our Turkic training corpora include Wikipedia dumps and the WikiANN evaluation set is itself derived from Wikipedia entity links (Pan et al., 2017). Entity-level overlap between training and test is therefore likely and systematically inflates NER F1 across all configurations that train on Wikipedia (E1, E2, E3, E4, E6). We do not quantify this overlap, which is a meaningful gap: overlap may be higher for Kazakh (largest Wikipedia corpus) than for Kyrgyz (curated non-Wikipedia sources), partially confounding cross-configuration NER comparisons. A non-Wikipedia NER benchmark such as KazNERD (Yeshpanov et al., 2022) would isolate

whether our transfer benefit reflects genuine entity learning or shared-entity memorization; we defer this to future work.

(L3) Rank-LR confound in E5 (addressed by E5b). Experiment E5 ($r=64$, $lr=5 \times 10^{-4}$) originally confounded rank and learning rate. The controlled E5b run ($r=64$, $lr=2 \times 10^{-4}$) resolves this: at matched rank, lowering the learning rate to baseline values eliminates the catastrophic collapse, pointing to learning rate as the proximate cause (see Section 4.1). The residual confound is now narrower: E5b still uses 5 epochs versus the baselines’ 3 epochs; a follow-up matching rank, LR, *and* training duration is desirable but did not fit within our compute budget.

(L4) Quantization confound. All experiments use 4-bit NF4 quantization, while the proposed SE threshold (Biderman et al., 2024) was established in full precision. Quantization noise may itself dampen singular value growth, so our “SE never exceeds 0.59” observation may reflect quantization as much as language-specific dynamics. A matched BF16 Kyrgyz baseline (which we are adding as a control in revision) would isolate whether SE behaves differently in full precision on the same underrepresented-language setting.

(L5) Corpus imbalance between Turkic languages (partially addressed by E1b). We match corpora by raw byte size (~ 150 MB each), yet token counts differ substantially after tokenization (KZ: 14.1M, UZ: 5.4M, KY: 4.4M). To isolate tokenizer-coverage effects from raw-data-volume effects, we ran a token-matched control E1b: the Kazakh corpus truncated to ~ 4.4 M tokens to match the Kyrgyz training set. Result: target KZ PPL rises from 2.73 (E1) to 4.17 (E1b) with the smaller corpus, confirming that data volume materially contributes to the E1 advantage. However, E1b’s 4.17 remains lower than the Kyrgyz baseline’s 4.78 at identical token count, so Kazakh also enjoys a genuine tokenizer-/pretraining-coverage advantage. Both factors contribute; neither alone explains the gap. A parallel control on Uzbek data would strengthen this conclusion; we defer it to future work.

(L6) NER as a primary metric. Our NER evaluation uses 3-shot prompting with $n=100$ samples per language and F1 derived

from parse-success output. This is closer to a test of instruction-following than of entity knowledge. The $r=64$ “NER F1 = 0” result, in particular, is driven by 100% parse failures; the model may retain some entity knowledge that this evaluation cannot access. Log-likelihood-based NER scoring (scoring candidate entity spans by model probability) would separate instruction loss from knowledge loss and is a priority follow-up.

(L7) Scope of generalization. All experiments use a single base model (Gemma-2-9B); results may differ for Llama 3, Mistral, or other architectures. We examine only three of the approximately 35 Turkic languages, excluding Turkish (the highest-resource Turkic language). Corpus size is fixed at ~ 150 MB per language, precluding scaling analyses. SVD monitoring focuses on `lora_B`; `lora_A` dynamics may reveal complementary patterns we do not capture.

(L8) Post-hoc Frobenius observations. The growth ranges in Section 4.4 are derived from the same experiments on which their interpretation is built; they are observations requiring held-out validation, not established thresholds, and we deliberately reframe them as such in Section 4.4.

7 Conclusion

We have presented a systematic study of the spectral dynamics of LoRA adapters during fine-tuning on three low-resource Turkic languages. Our experiments yield three principal findings.

First, the spectral energy threshold hypothesis is **not supported** in our setting: no experiment exceeds SE of 0.59 (well below the proposed 0.7 threshold), yet the high-LR $r=64$ configuration (E5) suffers complete functional collapse (NER F1 = 0, cross-lingual PPL > 600). Our E5/E5b control pair sharpens this falsification: at identical rank ($r=64$) and near-identical spectral trajectories (SE < 0.1 , Frobenius growth $15.6\times$ vs. $18.2\times$), only the high-LR run collapses—learning rate is the proximate cause, and spectral metrics cannot detect the difference. SE decreases during training as effective rank grows in both configurations, rendering it an unreliable diagnostic for training pathology.

Second, **Frobenius norm growth ratio** is a useful but language-dependent diagnostic. Within underrepresented languages, it reliably separates healthy from pathological regimes ($1.13\times$ for transfer vs. $15\text{--}38\times$ for $r=64$). However, our EN control reveals that well-represented languages can absorb comparable norm growth ($33\times$) with minimal forgetting, demonstrating that the *directionality* of weight perturbations—not just their magnitude—determines forgetting severity.

Third, **sequential cross-lingual transfer** between related Turkic languages robustly *retains source-language knowledge*: across two seeds, Kazakh $\rightarrow$ Kyrgyz transfer halves cross-lingual Kazakh PPL (23.5 ± 0.1 vs. 50.0 ± 2.4 under direct training) with minimal Frobenius norm growth ($+13\%$). A single-seed NER-F1 improvement ($+61\%$) proved not robust to reseeding, so we characterize the transfer effect primarily as retention rather than target-language improvement (Section D).

These findings have immediate practical implications for the growing community of researchers and practitioners adapting LLMs to low-resource languages. Future work should validate our findings across diverse model architectures (particularly in full-precision BF16 to control for quantization effects on spectral dynamics), extend the analysis to additional Turkic languages including Turkish, investigate directional norm regularization as a training objective, and explore adaptive rank selection strategies informed by real-time spectral monitoring. An open question is whether `lora_A` matrices exhibit complementary spectral signals when `lora_B` norms explode—joint analysis of both matrices may yield a more complete diagnostic.

Ethics Statement

This work involves fine-tuning language models on text corpora from three Turkic languages and English. While we aim to improve NLP capabilities for underserved language communities, we acknowledge that language models can amplify biases present in training data. Our training corpora, drawn from Wikipedia, literary texts, and web sources, may contain cultural biases or factual inaccuracies.

Data availability: The Kazakh and Uzbek corpora are compiled from publicly available sources (Wikipedia, open book collections). The Kyrgyz corpus is curated from licensed literary and encyclopedic sources and cannot be redistributed due to copyright restrictions. In the spirit of open science, we release all trained LoRA adapters, SVD monitoring logs, evaluation results, and full training code via GitHub and Hugging Face, enabling complete reproduction of the evaluation and analysis stages.

Acknowledgments

We thank the creators of the Kazakh, Kyrgyz, and Uzbek corpora used in this study, as well as the developers of the TUMLU benchmark for providing standardized Turkic language evaluation resources. All experiments were conducted on personal hardware without external funding.

Reproducibility Statement

All experimental code (training with SVD monitoring, evaluation, and plotting scripts), fine-tuned LoRA adapters for all eight experiments, SVD monitoring logs, and evaluation reports are publicly available at <https://github.com/Zarina33/Spectral-Collapse-Turkic> and <https://huggingface.co/Zarinaaa/spectral-collapse-turkic-adapters>. The Kazakh, Uzbek, and English training corpora are derived from publicly available sources and can be reconstructed using the provided data preparation scripts. The Kyrgyz corpus cannot be redistributed due to licensing restrictions on its source materials; consequently, the KY training stage is not fully reproducible. However, the released KY adapters and SVD logs enable full reproduction of all evaluation and analysis stages.

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A Detailed Training Configurations

All experiments were conducted on a single NVIDIA GeForce RTX 5080 (16 GB) using PyTorch 2.10, Transformers 5.2, and PEFT 0.18. Table 7 provides the full training statistics.

B Effective Rank Dynamics

Figure 8 shows the effective rank dynamics across experiments. The $r=64$ configuration shows increasing effective rank (from ~ 6 to ~ 26), indicating growing parameter utilization. The diverged experiment (E7) is the only configuration where effective rank *decreases*

ID	Experiment	Train Loss	Eval Loss	Eval PPL	Tokens	Time (min)
E1	KZ Baseline (14.1M)	0.275	1.259	3.52	14.1M	477
E1b	KZ Token-matched (4.4M)	1.186	1.461	4.31	4.4M	122
E2	UZ Baseline	1.193	1.822	6.18	5.4M	341
E3	KY Baseline	0.815	1.920	6.82	4.4M	183
E4	KY Overfit	1.050	1.963	7.12	4.4M	1193
E5	KY $r=64$ high-LR	0.274	2.164	8.71	4.4M	157
E5b	KY $r=64$ std-LR	1.278	1.964	7.13	4.4M	605
E6	KZ→KY	1.809	1.932	6.90	4.4M	355
E8	EN Control	1.821	2.017	7.51	12.7M	1351

Table 7: Full training statistics for all experiments. Eval loss and PPL are computed on the validation split during training (distinct from the post-hoc evaluation in Table 3). The higher train loss for E6 (KZ→KY) reflects the adapter starting from KZ-optimized weights that initially perform poorly on KY data.

(from 43.3 to 37.1), consistent with true spectral collapse.

E7 divergence analysis. The diverged experiment ($r=64$, $lr=10^{-3}$) provides the most extreme datapoint. Training terminated at step 800 due to loss divergence. Even at this point, maximum SE was only 0.18—still far below the 0.7 threshold—while mean Frobenius norm had already reached 6.75 ($19\times$ growth in just 800 steps, compared to $15.6\times$ over 4,885 steps for E5). The effective rank decrease ($43.3 \rightarrow 37.1$) in the final 200 steps is the only instance of true rank contraction in our experiments, suggesting that SE-based collapse detection may only trigger *after* the model has already diverged—too late for practical intervention. The Frobenius norm, by contrast, shows clear exponential growth from step 100 onward, providing a much earlier warning signal.

C Log-Likelihood NER: Type Classification under Gold Spans

To separate loss of entity knowledge from loss of instruction-following/output-formatting, we add a second NER evaluation that does not require the model to generate structured output. Given the gold-span boundaries from WikiANN, we present each span in context and score three candidate completions (“person”, “organization”, “location”) by log-likelihood under the adapter; the predicted type is the argmax. Because this evaluation never relies on the model’s output format, it is immune to the 100% parse-failure mode that drives E5’s NER $F1 = 0$ under generation-based scoring.

Table 8 reports per-language type accuracy and macro-F1 on the same 100-example test sets used for Table 3.

D Seed-Variance Replication (E3 and E6)

To provide at least partial seed-variance estimates for our most consequential configurations, we rerun two experiments under a second seed (123, while the originals used seed 42). Training hyperparameters and data are identical across seeds; only the random seed for initialization, data shuffling, and train/val split differs.

These results sharpen the interpretive stance taken throughout the paper: we treat cross-lingual PPL differences as the most reliable quantitative signal, and the various NER scores as directional indicators whose per-experiment magnitudes should not be over-interpreted. A full $n=3$ replication of all eight experiments remains the most important reproducibility follow-up (see L1); the present two-seed data covers the two configurations whose comparison underlies our main transfer claim.

E Per-Type NER Results

Table 10 breaks down NER performance by entity type for the Kyrgyz evaluation.

F TUMLU Accuracy by Language

Table 11 presents TUMLU accuracy scores across all languages and experiments.

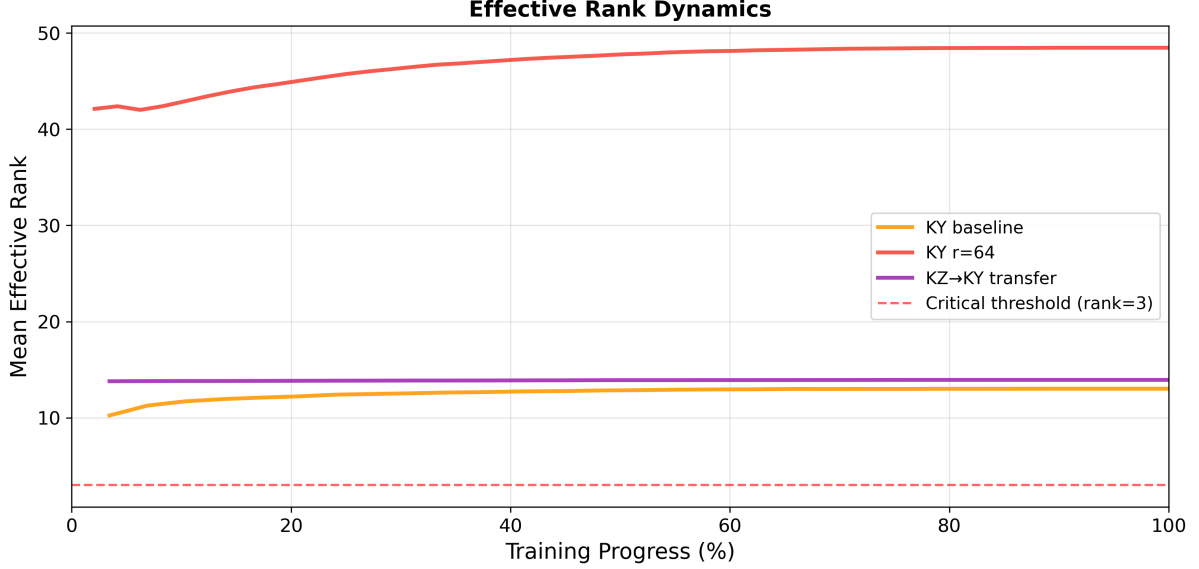


Figure 8: Effective rank dynamics during training. The $r=64$ configuration shows increasing effective rank despite functional degradation, further evidence that standard spectral metrics can be misleading.

G Functional Collapse: Raw Output Examples

To illustrate the nature of functional collapse at $r=64$, we present raw model outputs for two tasks. The model completely ignores instructions and exhibits degenerate repetition.

NER prompt: Extract named entities from the text. Format each as: ENTITY ||| TYPE

Input: Бишкекте Кыргыз Республикасынын Президенти Садыр Жапаров менен Казакстандын Президенти Касым-Жомарт Токаев жолукту.

$r=64$ **output:** Кыргыз Республикасынын Президентинин кызматына кандидат болгон. Кыргыз Республикасынын Президентинин кызматына кандидат болгон. Кыргыз Республикасынын Президентинин кызматына кандидат болгон... (repeats indefinitely)

Text continuation prompt: Кыргызстан --- это

$r=64$ **output:** не только, не только, не только, не только... (single bigram repeated 25+ times until max tokens)

Both examples demonstrate that the model has lost instruction-following capability entirely. Rather than extracting entities or generating coherent text, it produces degenerate

repetitions—a hallmark of functional collapse distinct from spectral collapse.

Experiment	KY	KZ	UZ
E1 KZ Baseline (14.1M tok)	57.7%	72.3%	70.9%
E1b KZ Token-matched (4.4M)	46.0%	70.5%	69.9%
E2 UZ Baseline	54.9%	71.4%	65.0%
E3 KY Baseline	50.4%	69.6%	63.1%
E4 KY Overfit (10ep)	56.8%	71.4%	60.2%
E5 KY $r=64$ (lr 5×10^{-4})	59.5%	50.0%	55.3%
E5b KY $r=64$ (lr 2×10^{-4})	53.1%	70.5%	69.9%
E6 KZ→KY	59.5%	73.2%	66.0%
E8 EN Control	56.8%	73.2%	68.0%

Table 8: Log-likelihood span-type accuracy across experiments (gold-span boundaries, argmax over {PER, ORG, LOC}). Random-baseline accuracy is 33.3%. The parse-failure-immune evaluation reveals that **E5 retains and even marginally exceeds target-language (KY) entity knowledge** (59.5%, tied for best with E6) despite scoring $F1 = 0$ under generation-based NER—confirming that the high-LR $r=64$ collapse destroys instruction-following, not entity knowledge itself. Cross-lingual degradation, however, does appear in E5 on non-target languages (KZ 50.0% vs. 70–73% for others), consistent with the cross-lingual PPL increase. Under this cleaner metric, the KZ→KY transfer advantage over the KY baseline is modest (59.5% vs. 50.4% on KY; +9.1 absolute percentage points) rather than the +61% relative improvement suggested by generation F1, reinforcing our framing of cross-lingual transfer as a *directional* rather than precise effect.

	E3 (KY baseline)		E6 (KZ→KY)		E5b ($r=64$, $\text{lr } 2 \times 10^{-4}$)	
Metric	s=42	s=123	s=42	s=123	s=42	s=123
KY PPL ↓	4.78	4.83	4.73	4.78	4.55	4.62
KZ PPL ↓	47.58	52.47	23.49	23.61	128.04	107.23
UZ PPL ↓	56.85	54.05	124.19	114.54	162.68	139.89
NER F1 KY (gen) ↑	0.157	0.182	0.253	0.179	0.146	0.116
NER F1 KZ (gen) ↑	0.190	0.186	0.130	0.185	0.129	0.126
NER TypeAcc KY ↑	50.4%	59.5%	59.5%	64.9%	53.1%	64.9%
NER TypeAcc KZ ↑	69.6%	70.5%	73.2%	75.0%	70.5%	71.4%
TUMLU KY ↑	34.7%	37.4%	33.7%	34.9%	33.2%	33.2%

Table 9: Two-seed replication for the three configurations underlying our main claims: KY baseline (E3), KZ→KY transfer (E6), and the rank-LR control (E5b). Target-language PPL is highly seed-stable across all three (± 0.05 – 0.07). NER F1 under generation carries the largest variance (± 0.02 – 0.04), and log-likelihood span-typing accuracy on KY shows ± 5 – 9 pp seed variance. Two robust claims emerge: **(i)** the halving of cross-lingual Kazakh PPL under transfer (23.5 ± 0.1 vs. 50.0 ± 2.4 under direct training), and **(ii)** the disentanglement of learning rate from rank in collapse onset: both E5b seeds sustain NER F1 ≥ 0.12 , TUMLU accuracy of 33.2%, and KY PPL ≤ 4.62 , all well outside the collapsed regime of E5 (F1 = 0.0, TUMLU 23.2%, KY PPL 5.9). The NER F1 +61% transfer headline at seed 42 drops to -2% at seed 123; we therefore interpret transfer as primarily a retention mechanism.

Experiment	PER F1	ORG F1	LOC F1
KZ Baseline 14.1M (on KY)	0.236	0.162	0.229
KZ Token-matched 4.4M (on KY)	0.101	0.207	0.261
UZ Baseline (on KY)	0.095	0.150	0.209
KY Baseline	0.101	0.160	0.266
KY Overfit	0.224	0.255	0.119
KY $r=64$ (E5, $\text{lr } 5 \times 10^{-4}$)	0.000	0.000	0.000
KY $r=64$ (E5b, $\text{lr } 2 \times 10^{-4}$)	0.126	0.178	0.177
KZ→KY	0.344	0.202	0.214

Table 10: Per-type NER F1 scores on Kyrgyz WikiANN. The KZ→KY transfer achieves the highest PER F1 (0.344), substantially outperforming the KY baseline (0.101). The $r=64$ configuration produces zero scores across all types due to 100% parse failures.

Experiment	KY	KZ	UZ
KZ Baseline (14.1M tok)	35.7%	38.4%	32.8%
KZ Token-matched (4.4M tok)	37.3%	35.6%	32.8%
UZ Baseline	31.9%	33.1%	35.4%
KY Baseline	34.7%	30.2%	29.6%
KY Overfit	33.9%	27.8%	31.5%
KY $r=64$ (E5, $\text{lr } 5 \times 10^{-4}$)	23.2%	22.2%	22.9%
KY $r=64$ (E5b, $\text{lr } 2 \times 10^{-4}$)	33.2%	27.7%	26.6%
KZ→KY	33.7%	31.7%	30.8%

Table 11: TUMLU accuracy across all experiments and languages. The KZ baseline achieves the highest accuracy on its target language (38.4%). The $r=64$ configuration drops to near-random levels ($\sim 23\%$ on 4-option multiple choice) across all languages.